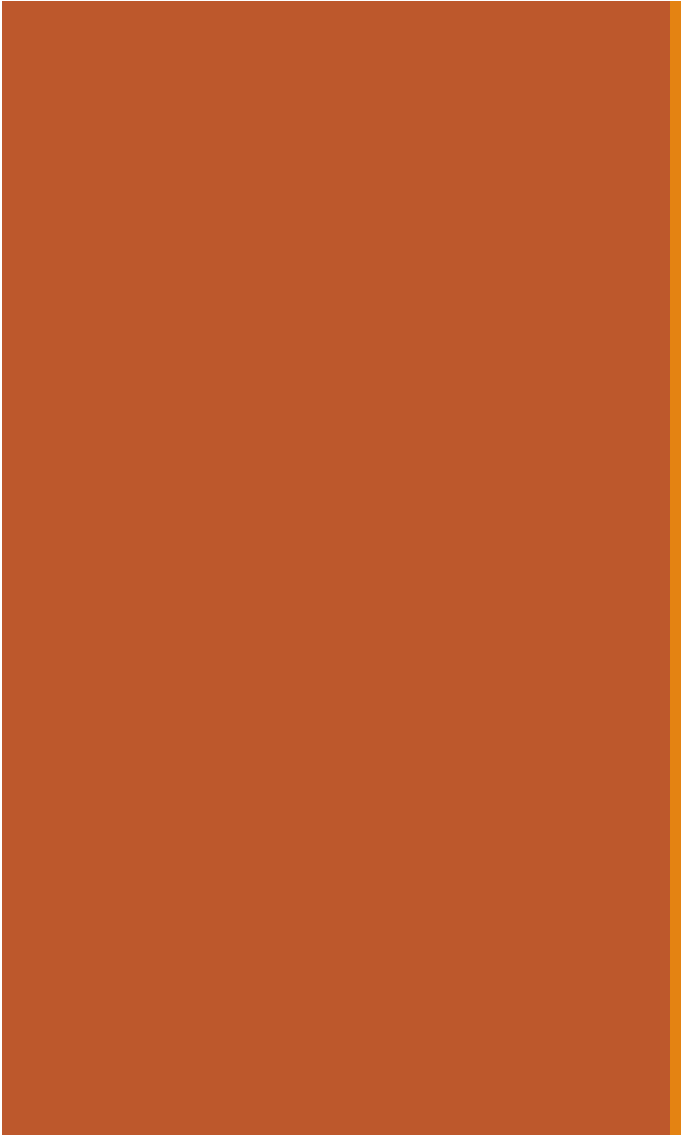


14: Conditional Expectation

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[Lecture Discussion on Ed](#)





Discrete conditional distributions

Discrete conditional distributions

Recall the definition of the conditional probability of event E given event F :

$$P(E|F) = \frac{P(EF)}{P(F)}$$

For discrete random variables X and Y , the **conditional PMF** of X given Y is

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

*this is essentially
saying the same thing,
since $X=x$ and
 $Y=y$ are events.
 $\equiv$*

Different notation,
same idea:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

Discrete probabilities of CS109

must be that $\sum_{t=1}^1 \sum_{y=1}^3 p(y,t) = 1$

Each student responds with:

Year Y

- 1: Freshmen and Sophomores
- 2: Juniors and Seniors
- 3: Graduate Students and SCPD

Mood T :

- -1: 😞
- 0: 😐
- 1: 😍

	Joint PMF		
	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

this row is concerned with just those people who are feeling meh 😐

$P(Y = 2, T = 1)$

this column is only concerned with the world of sophomores and freshmen.

Joint PMFs sum to 1.

Discrete probabilities of CS109

The below are **conditional probability tables** for conditional PMFs

(A) $P(Y = y|T = t)$ and (B) $P(T = t|Y = y)$.

- Which is which?
- What's the missing probability?

	Joint PMF		
	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.09	.04	.08
$T = 0$	.45	.61	.75
$T = 1$	.46	.35	.17

each column sums to 1.

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.75	.125	?
$T = 0$	.56	.27	.17
$T = 1$	.75	.2	.05

these two rows add up to 1.



Discrete probabilities of CS109

The below are **conditional probability tables** for conditional PMFs

(A) $P(Y = y|T = t)$ and (B) $P(T = t|Y = y)$.

- Which is which?
- What's the missing probability?

	Joint PMF		
	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

(B) $P(T = t|Y = y)$

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.09	.04	.08
$T = 0$	.45	.61	.75
$T = 1$	.46	.35	.17

$.30 / (.06 + .29 + .30)$
how does T distribute itself when Y=2?

(A) $P(Y = y|T = t)$

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.75	.125	.125
$T = 0$	.56	.27	.17
$T = 1$	.75	.2	.05

how does Y distribute itself when T=1?

Conditional PMFs also sum to 1 conditioned on different events!

Quick check

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Number or function?

1. $P(X = 2|Y = 5)$

2. $P(X = x|Y = 5)$

3. $P(X = 2|Y = y)$

4. $P(X = x|Y = y)$

True or false?

5. $\sum_x P(X = x|Y = 5) = 1$

6. $\sum_y P(X = 2|Y = y) = 1$

7. $\sum_x \sum_y P(X = x|Y = y) = 1$

8. $\sum_x \left(\sum_y P(X = x|Y = y) P(Y = y) \right) = 1$



Quick check

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Number or function?

True or false?

1. $P(X = 2|Y = 5)$

number

2. $P(X = x|Y = 5) = \frac{P_{X,Y}(x,5)}{P_Y(5)}$

1-D function

3. $P(X = 2|Y = y) = \frac{P_{X,Y}(2,y)}{P_Y(y)}$

1-D function

4. $P(X = x|Y = y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$

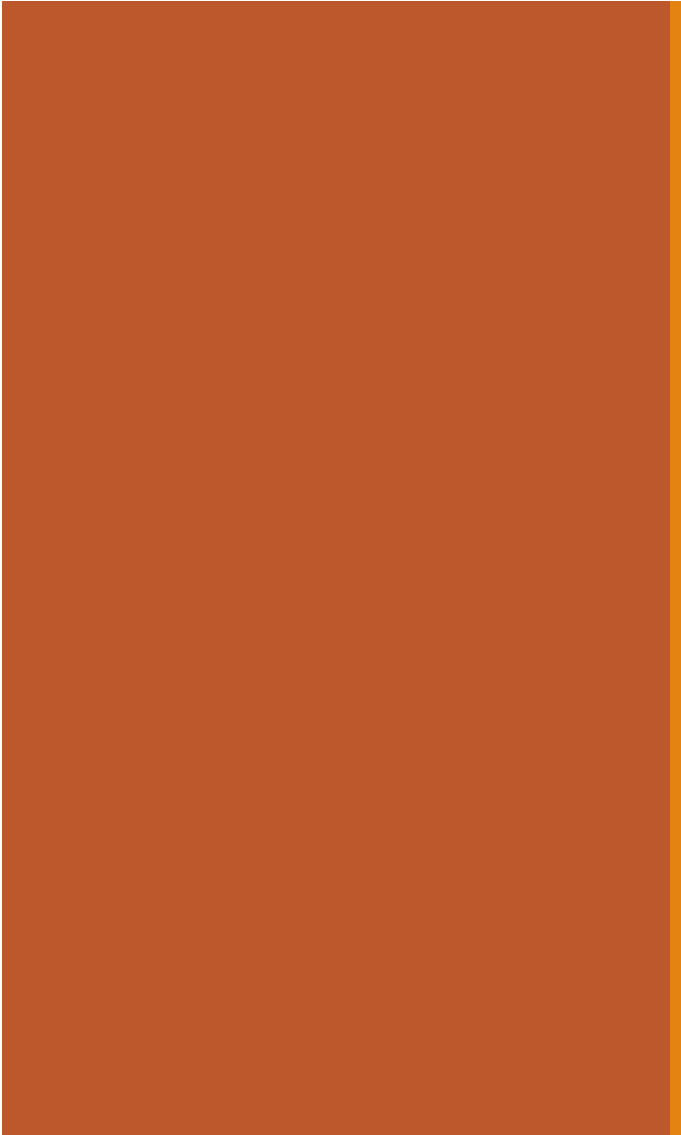
2-D function

5. $\sum_x P(X = x|Y = 5) = 1$ true

6. $\sum_y P(X = 2|Y = y) = 1$ false

7. $\sum_x \sum_y P(X = x|Y = y) = 1$ false

8. $\sum_x \left(\sum_y \underbrace{P(X = x|Y = y)P(Y = y)}_{\text{by the chain rule, this is just } P(X=x, Y=y)} \right) = 1$ true



Conditional Expectation

Conditional expectation

Recall the conditional PMF of X given $Y = y$:

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

The **conditional expectation** of X given $Y = y$ is

$$E[X|Y = y] = \sum_x xP(X = x|Y = y) = \sum_x xp_{X|Y}(x|y)$$

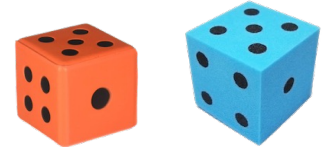
$Y=y$
is the new
sample
space.

- Note that $E[X]$ is a well-defined statistic even when X is one of many random variables in a multivariate distribution: $E[X] = \sum_x \sum_y xp_{X,Y}(x, y)$
- $E[X|Y = y]$ is the average value of X when Y is constrained to take on a specific value of y : $E[X|Y = y] = \sum_x xp_{X,Y}(x|y)$

It's been so long, our dice friends

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Roll two 6-sided dice.
- Let roll 1 be D_1 , roll 2 be D_2 .
- Let S = value of $D_1 + D_2$.



note that $S|D_2=6$ is a random variable!

1. What is $E[S|D_2 = 6]$?
$$E[S|D_2 = 6] = \sum_x x P(S = x|D_2 = 6)$$
$$= \left(\frac{1}{6}\right) (7 + 8 + 9 + 10 + 11 + 12)$$
$$= \frac{57}{6} = 9.5$$

Intuitively: $6 + E[D_1] = 6 + 3.5 = 9.5$ We'll prove in a moment

Properties of conditional expectation

1. LOTUS:

$$E[g(X)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$$

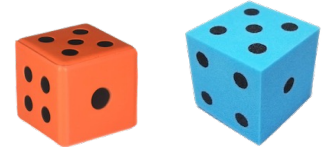
2. Linearity of conditional expectation:

$$E\left[\sum_{i=1}^n X_i \mid Y = y\right] = \sum_{i=1}^n E[X_i | Y = y]$$

3. Law of total expectation (in, like, three slides)

It's been so long, our dice friends

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$



- Roll two 6-sided dice.
- Let roll 1 be D_1 , roll 2 be D_2 .
- Let $S = \text{value of } D_1 + D_2$.

1. What is $E[S|D_2 = 6]$?

$$\frac{57}{6} = 9.5$$

2. What is $E[S|D_2]$?

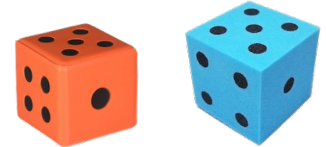
- A. A function of S
- B. A function of D_2
- C. A number

3. Give an expression for $E[S|D_2]$.



It's been so long, our dice friends

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$



- Roll two 6-sided dice.
- Let roll 1 be D_1 , roll 2 be D_2 .
- Let $S = \text{value of } D_1 + D_2$.

1. What is $E[S|D_2 = 6]$?

$$\frac{57}{6} = 9.5$$

2. What is $E[S|D_2]$?

- A. A function of S
 - B. A function of D_2**
 - C. A number
3. Give an expression for $E[S|D_2]$.

$$E[S|D_2 = d_2] = E[D_1 + d_2|D_2 = d_2]$$

$$= \sum_{d_1} (d_1 + d_2) P(D_1 = d_1 | D_2 = d_2)$$

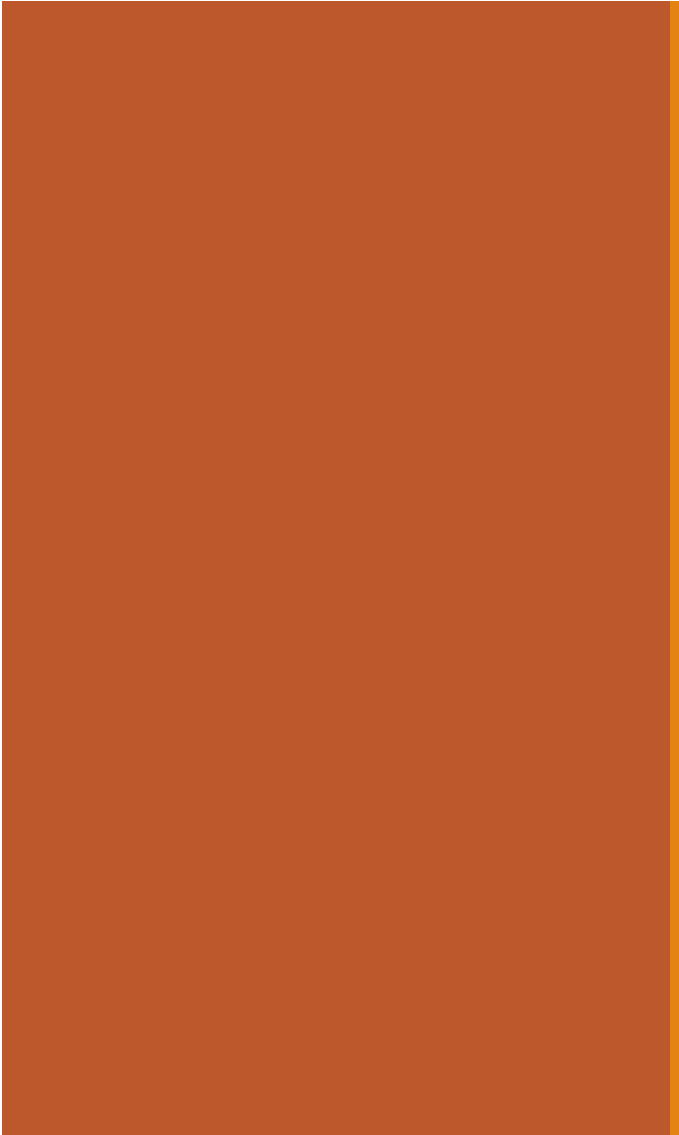
$$= \sum_{d_1} d_1 P(D_1 = d_1) + d_2 \sum_{d_1} P(D_1 = d_1)$$

($D_1 = d_1, D_2 = d_2$ independent events)

definition of $E[D_1]$ *this sums to 1*

$$= E[D_1] + d_2 = 3.5 + d_2$$

$$E[S|D_2] = 3.5 + D_2$$



Law of Total Expectation

Properties of conditional expectation

1. LOTUS:

$$E[g(X)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$$

2. Linearity of conditional expectation:

$$E\left[\sum_{i=1}^n X_i \mid Y = y\right] = \sum_{i=1}^n E[X_i | Y = y]$$

3. Law of total expectation:

$$E[X] = E[E[X|Y]] \quad \text{what?}$$

this inner expectation
is a random variable
on Y .

Proof of Law of Total Expectation

$$E[X] = E[E[X|Y]]$$

$$\begin{aligned}
 E[E[X|Y]] &= E[g(Y)] = \sum_y P(Y = y) E[X|Y = y] && \text{(LOTUS, } g(Y) = E[X|Y]) \\
 &= \sum_y P(Y = y) \sum_x x P(X = x|Y = y) && \begin{array}{l} \text{this is the definition of } E[X|Y=y] \\ \text{(def of conditional expectation)} \end{array} \\
 &= \sum_y \left(\sum_x x P(X = x|Y = y) P(Y = y) \right) = \sum_y \left(\sum_x x P(X = x, Y = y) \right) && \begin{array}{l} \text{this is a constant with respect to } x \\ \text{(chain rule)} \end{array} \\
 &= \sum_x \sum_y x P(X = x, Y = y) = \sum_x x \sum_y P(X = x, Y = y) && \begin{array}{l} \text{switch order of summations} \\ \text{(marginalization)} \end{array} \\
 &= \sum_x x P(X = x) && \text{this defines the marginal probability of } X!! \\
 &= E[X]
 \end{aligned}$$

Another way to compute $E[X]$

$$E[X] = E[E[X|Y]]$$

$$E[E[X|Y]] = \sum_y P(Y = y)E[X|Y = y] = E[X]$$

If we only have a conditional PMF of X on some discrete variable Y , we can compute $E[X]$ as follows:

1. Compute expectation of X given some value of $Y = y$
2. Repeat step 1 for all values of Y
3. Compute a weighted sum (where weights are $P(Y = y)$)

```
def recurse():  
    if random.random() < 0.5:  
        return 3  
    return 2 + recurse()
```

Useful for analyzing recursive code.

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y]P(Y = y)$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    if x == 2: return 5 + recurse()  
    return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y]P(Y = y)$$

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    return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X = 1]P(X = 1) + E[Y|X = 2]P(X = 2) + E[Y|X = 3]P(X = 3)$$



$$E[Y|X = 1] = 3$$

When $X = 1$, return 3.

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    if x == 2: return 5 + recurse()  
    return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

↑
 $E[Y|X=1] = 3$

What is $E[Y|X=2]$?

- A. $E[5] + Y$
- B. $E[5 + Y] = 5 + E[Y]$
- C. $5 + E[Y|X=2]$



Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
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    return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$


 $E[Y|X=1] = 3$

 When $X=2$, return 5 +
a future return value of `recurse()`.

What is $E[Y|X=2]$?

- A. $E[5] + Y$
- B. $E[5 + Y] = 5 + E[Y]$
- C. $5 + E[Y|X=2]$

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    if x == 2: return 5 + recurse()  
    return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$


 $E[Y|X=1] = 3$


 $E[Y|X=2] = 5 + E[5 + Y]$

 When $X=3$, return
7 + a future return value
of `recurse()`.

$$E[Y|X=3] = E[7 + Y]$$

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
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Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ E[Y|X=1] = 3 & E[Y|X=2] = 5 + E[Y] & E[Y|X=3] = 7 + E[Y] \end{array}$$

$$E[Y] = 3(1/3) + (5 + E[Y])(1/3) + (7 + E[Y])(1/3)$$

$$E[Y] = (1/3)(15 + 2E[Y]) = 5 + (2/3)E[Y]$$

$$E[Y] = 15$$

On your own: What is $\text{Var}(Y)$?

Independent RVs, defined another way

If X and Y are **independent** discrete random variables, then $\forall x, y$:

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P(X = x)P(Y = y)}{P(Y = y)} = P(X = x)$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x)$$

Note for conditional expectation, independent X and Y implies

$$E[X|Y = y] = \sum_x xp_{X|Y}(x|y) = \sum_x xp_X(x) = E[X]$$

Random number of random variables

$$\begin{array}{l} \text{indep } X, Y \\ E[X|Y = y] = E[X] \end{array}$$

Suppose you have a website: **doristhebeast.com**. Let:

- X = # of people per day who visit your site. $X \sim \text{Poi}(50)$
- Y_i = # of minutes spent per day by visitor i $Y_i \sim \text{Poi}(11)$
- X and all Y_i are independent.

The time spent by all visitors per day is $W = \sum_{i=1}^X Y_i$. What is $E[W]$?



Random number of random variables

Suppose you have a website: **dorisisthebeast.com**. Let:

- X = # of people per day who visit your site. $X \sim \text{Poi}(50)$
- Y_i = # of minutes spent per day by visitor i . $Y_i \sim \text{Poi}(11)$
- X and all Y_i are independent.

The time spent by all visitors per day is $W = \sum_{i=1}^X Y_i$. What is $E[W]$?

$$E[W] = E\left[\sum_{i=1}^X Y_i\right] = E\left[E\left[\sum_{i=1}^X Y_i \mid X\right]\right]$$

this really is $E[W|X]$

$$= E[XE[Y_i]]$$

$$= E[Y_i]E[X] \quad (\text{scalar } E[Y_i])$$

$$= 11 \cdot 50 = 550$$

Suppose $X = x$.

$$E\left[\sum_{i=1}^x Y_i \mid X = x\right] = \sum_{i=1}^x E[Y_i \mid X = x] \quad (\text{linearity})$$

$$= \sum_{i=1}^x E[Y_i] \quad (\text{independence})$$

$$= xE[Y_i]$$